

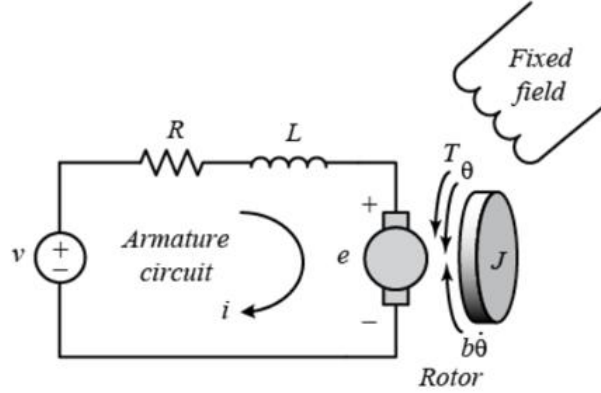


Figure 2 Circuit Diagram of DC Motor

Governing Differential Equation

1. The Electrical System (KVL)

Applying Kirchhoff's Voltage Law (KVL) to the armature circuit loop:

$$V(t) = V_R(t) + V_L(t) + e(t)$$

Where:

- $V_R = i(t)R$ (Voltage drop across armature resistance)
- $V_L = L \frac{di(t)}{dt}$ (Voltage drop across armature inductance)
- $e(t) = K_e \omega(t) = K_e \frac{d\theta(t)}{dt}$ (Back EMF produced by the rotating coil)

Substituting these values into the KVL equation yields **Equation (1)**:

$$L \frac{di(t)}{dt} + Ri(t) = V(t) - K_e \frac{d\theta(t)}{dt}$$

2. The Mechanical System (Newton's 2nd Law)

The sum of torques acting on the rotor inertia must equal its angular acceleration:

$$\sum T = J\alpha(t) = J \frac{d\omega(t)}{dt} = J \frac{d^2\theta(t)}{dt^2}$$

The acting torques are:

- **Driving Motor Torque (T_m):** Proportional to armature current, $T_m = K_t i(t)$
- **Opposing Friction Torque (T_f):** Due to bearing viscous damping, $T_f = b\omega(t) = b \frac{d\theta(t)}{dt}$

Setting up the balance equation:

$$T_{\text{driving}} - T_{\text{opposing}} = J \frac{d\omega(t)}{dt}$$

$$K_t i(t) - b\omega(t) = J \frac{d\omega(t)}{dt}$$

Rearranging gives us **Equation (2)** in terms of angular velocity (ω):

$$J \frac{d\omega(t)}{dt} + b\omega(t) = K_t i(t)$$

Or substituting $\omega(t) = \frac{d\theta(t)}{dt}$ to express it in terms of angular position (θ):

$$J \frac{d^2\theta(t)}{dt^2} + b \frac{d\theta(t)}{dt} = K_t i(t)$$

Transfer Function Development

Transforming to the Laplace Domain (s -Domain)

Assuming **zero initial conditions** ($i(0) = 0, \theta(0) = 0, \dot{\theta}(0) = 0$), we apply the Laplace Transform ($\frac{d}{dt} \rightarrow s$):

From the **Electrical Equation (1)**:

$$LsI(s) + RI(s) = V(s) - K_e s\Theta(s)$$

$$(Ls + R)I(s) = V(s) - K_e s\Theta(s) \quad \text{--- (3)}$$

From the **Mechanical Equation (2)** (using the position form):

$$Js^2\Theta(s) + bs\Theta(s) = K_t I(s)$$

$$s(Js + b)\Theta(s) = K_t I(s)$$

Isolate current $I(s)$ so we can substitute it out:

$$I(s) = \frac{s(Js + b)\Theta(s)}{K_t} \quad \text{--- (4)}$$

Algebraic Substitution & Simplification

Substitute Equation (4) into Equation (3):

$$(Ls + R) \left[\frac{s(Js + b)\Theta(s)}{K_t} \right] = V(s) - K_e s\Theta(s)$$

To eliminate the fraction, multiply the entire equation by K_t :

$$s(Js + b)(Ls + R)\Theta(s) = K_t V(s) - K_t K_e s\Theta(s)$$

Move all terms involving $\Theta(s)$ to the left-hand side:

$$s(Js + b)(Ls + R)\Theta(s) + K_t K_e s\Theta(s) = K_t V(s)$$

Factor out $s\Theta(s)$ on the left side:

$$[(Js + b)(Ls + R) + K_t K_e]s\Theta(s) = K_t V(s)$$

Final Transfer Functions:

A. Angular Position Transfer Function: $\frac{\Theta(s)}{V(s)}$

Isolating $\Theta(s)$ relative to input voltage $V(s)$:

$$\frac{\Theta(s)}{V(s)} = \frac{K_t}{s[(Js + b)(Ls + R) + K_t K_e]}$$

B. Angular Velocity Transfer Function: $\frac{\Omega(s)}{V(s)}$

Since angular velocity is the derivative of position, its Laplace relationship is $\Omega(s) = s\Theta(s)$.

Substituting this directly into our last factored step gives:

$$[(Js + b)(Ls + R) + K_t K_e]\Omega(s) = K_t V(s)$$

Dividing both sides yields the velocity transfer function:

$$\frac{\Omega(s)}{V(s)} = \frac{K_t}{(Js + b)(Ls + R) + K_t K_e}$$

$$\frac{\Omega(s)}{V(s)} = \frac{K_t}{JLs^2 + (JR + bL)s + (bR + K_t K_e)}$$